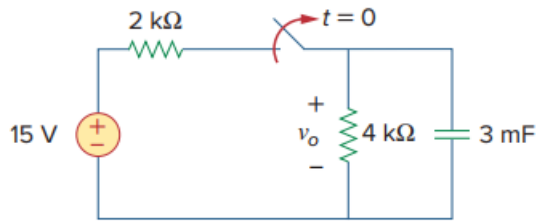


Problem

The switch in Fig. 7.89 opens at $t = 0$. Find v_o for $t > 0$.

Fig. 7.89:

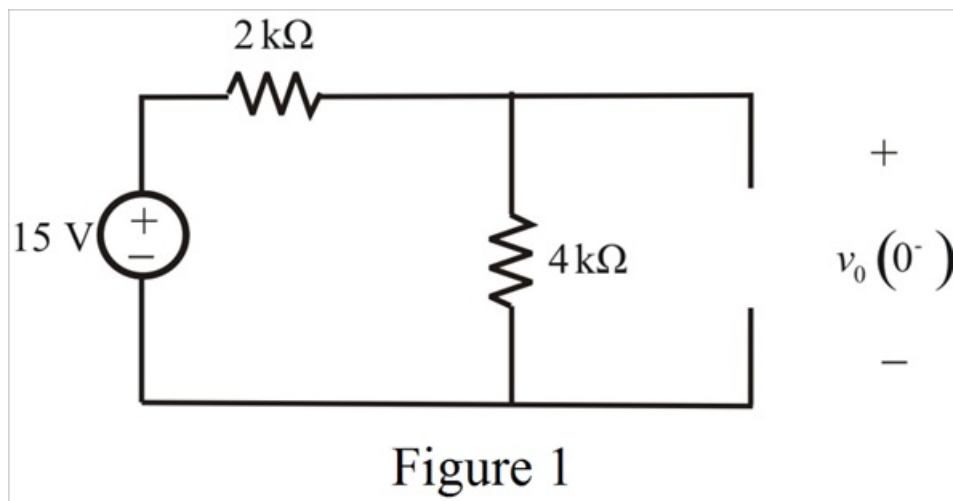


Step-by-step solution

Step 1 of 4

Consider that the switch opens at $t = 0$ and before that, it was closed for a long time. Capacitor acts like open circuit to DC.

Draw the circuit before $t = 0$ as shown in figure 1.



Step 2 of 4

By voltage division rule

$$v_o(0^-) = 15 \times \frac{4k}{4k + 2k}$$

$$v_o(0^-) = 10 \text{ V}$$

Capacitor doesn't allow sudden changes in the voltage. Therefore,

$$v_o(0^-) = v_o(0) = 10 \text{ V}$$

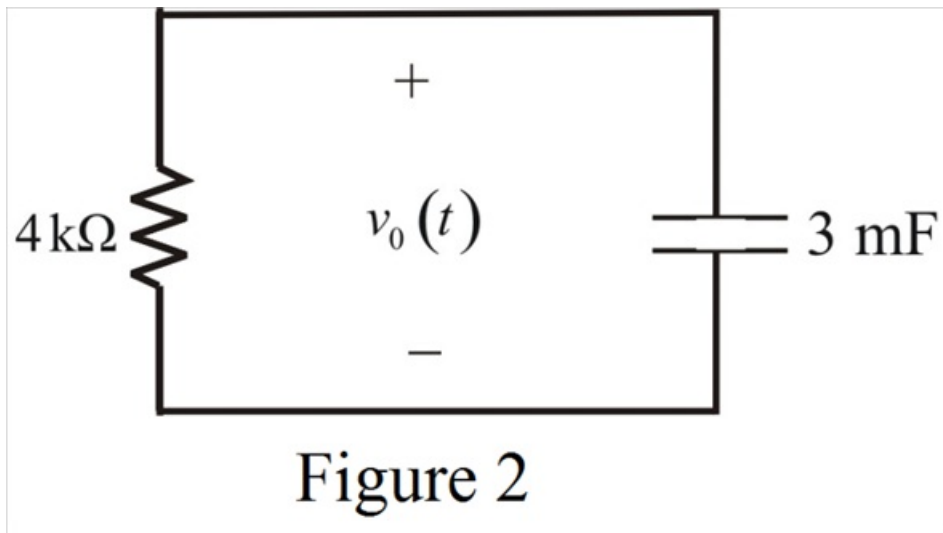
Consider that the switch opens at $t = 0$. For $t > 0$ circuit is source free network. Therefore, the voltage across the capacitor is

$$v_o(t) = v(0)e^{-\frac{t}{\tau}}$$

$$v_o(t) = 10e^{-\frac{t}{\tau}}$$

Step 3 of 4

Draw the circuit for $t > 0$ as shown in figure 2.



Step 4 of 4

Calculate the time constant.

$$\tau = RC$$

$$\tau = 4 \times 10^3 \times 3 \times 10^{-3}$$

$$\tau = 12 \text{ sec}$$

$$v_0(t) = 10e^{-\frac{t}{12}} \text{ V}$$

$$v_0(t) = 10e^{-0.08333t} \text{ V}$$

Therefore, the value of voltage $v_0(t)$ for $t > 0$ is $10e^{-0.08333t} \text{ V}$.